

DSAA 4011 · LAB 5 · SATURDAY 10 OCTOBER 2026

Learning What We Never See

Expectation–Maximization at the Frontier

Lecture 5's **expectation–maximization** (EM): **guess** what is hidden, **count**, repeat. Today: five places where this simple loop runs modern AI.

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Builds on Lecture 5: Learning what we never observe

Outline

- ▶ **Learning what we never see (Lecture 5)**

Troubles 1–2: cleaning and searching data

Trouble 3: inside image models

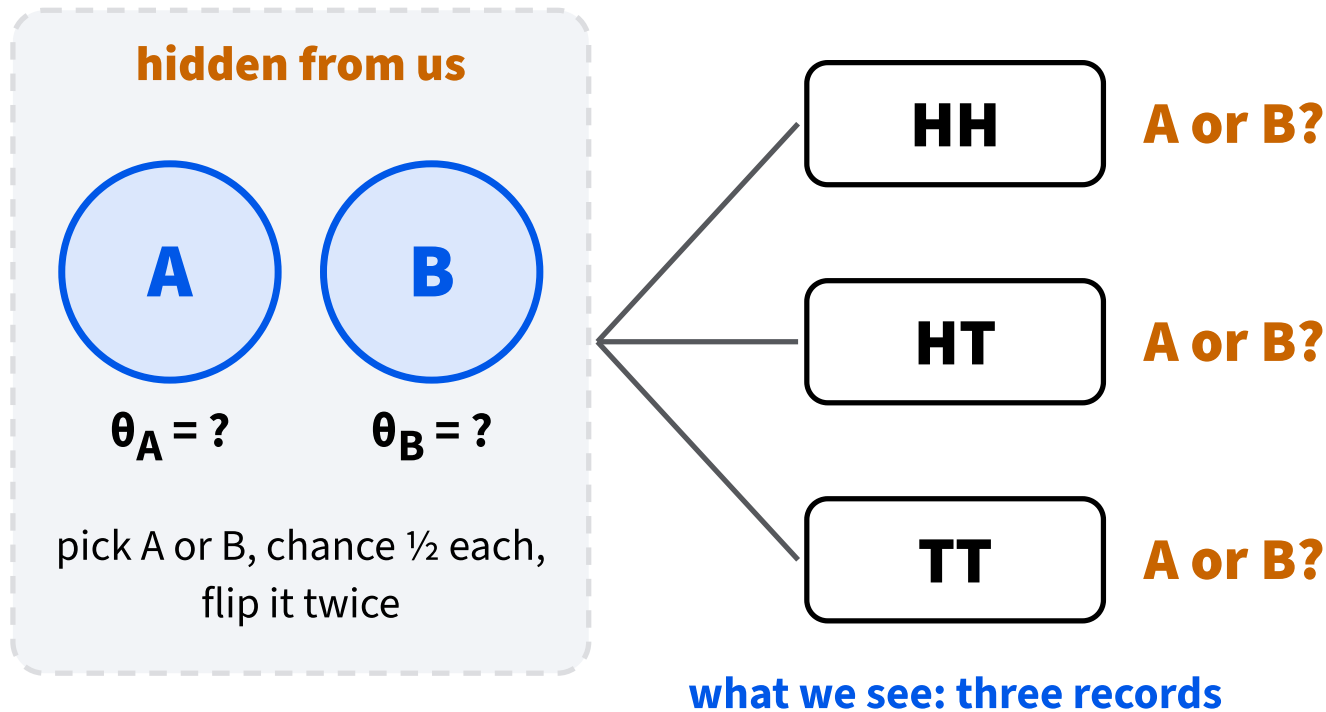
Conclusion and your turn

After class: Programs 4–5

One loop, five troubles: three today, two as programs.

Learning what we never see

Two coins, one hidden: which coin made each record?



Look: we see the flips, **never the coin**.

Q: Can we learn θ_A and θ_B anyway?

A: Yes: guess the coin **softly**, count heads, repeat.

Learning what we never see: in three sentences

BACKGROUND Data hide part of the story: which coin, which label, which reasoning.

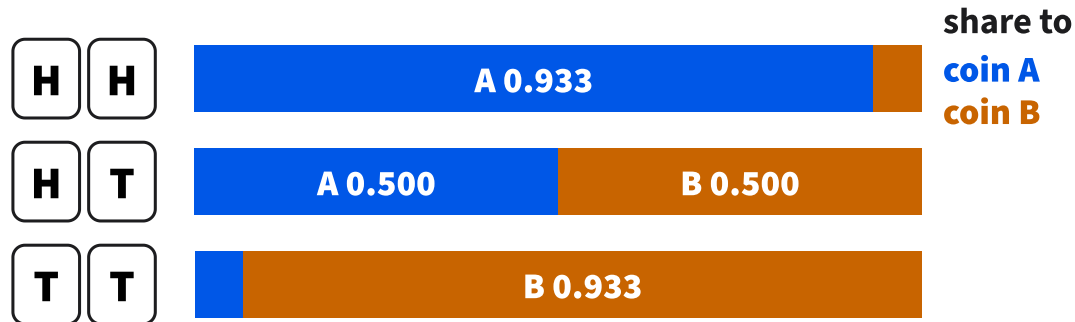
KEY IDEA **Guess** the hidden part softly (**E-step**), then **count** (**M-step**). Repeat.

USED FOR Every model with a hidden part [Dempster, Laird and Rubin, 1977].

Learning what we never see

Guess **softly**, **count**, repeat.

round 15 · M-step: count



heads and flips, credited by the soft labels:

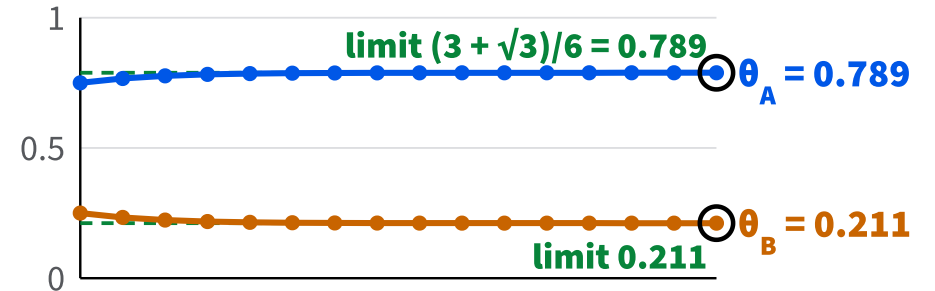
heads to A: $2 \cdot 0.933 + 1 \cdot 0.500 + 0 \cdot 0.067 = 2.366$

flips to A: $2 \cdot (0.933 + 0.500 + 0.067) = 3.000$

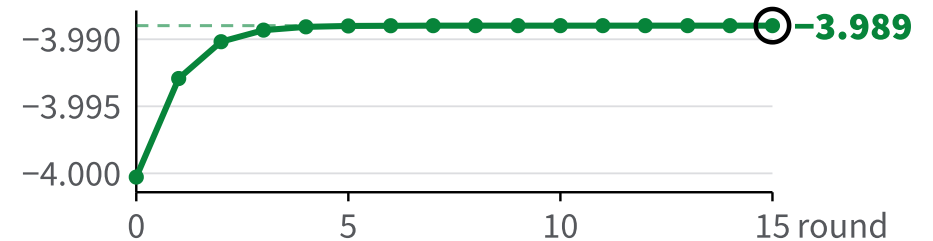
$\theta_A = 2.366 \div 3.000 = 0.789$

$\theta_B = 0.634 \div 3.000 = 0.211$

heads chances



log-likelihood $\ln p(\text{HH}, \text{HT}, \text{TT})$: never goes down



Watch: HH looks like coin A: **soft label** 0.9 (Lab 5, p. 60). Round by round θ_A climbs to **0.789**, and the fit (likelihood) **never drops**.

Q: Why not pick one coin per record? **A:** HT is 50 : 50; a hard pick is far too sure.

Try: another start.

Main Question

**EM only guesses and counts.
Can it judge a crowd, search a library, read images, cut
words, and teach reasoning?**

Yes: one small change each.

Learning what we never see

Five words for today.

Word	Plain meaning	Lecture 5 name
record x	what we see	observed data
hidden coin z	what we never see	hidden variable
soft label	how much we believe each z	posterior, E-step
count	heads weighted by soft labels	M-step
likelihood	chance of all records; never drops	Theorem 4 (Lecture 5, p. 49)

Learning what we never see

Five troubles at the frontier, five fixes.

	Trouble	Fix	Used for
1	people disagree on labels	???	clean training labels
2	too much data	???	search for chatbots
3	the hidden part is many numbers	???	image models
4	nobody marks where to cut words	???	tokenizers · program
5	the reasoning is never written	???	reasoning models · program

Read: the fixes appear one by one.

Outline

Learning what we never see (Lecture 5)

- ▶ **Troubles 1–2: cleaning and searching data**

Trouble 3: inside image models

Conclusion and your turn

After class: Programs 4–5

First: clean the labels, then search the data.

Trouble 1 in three sentences

- BACKGROUND** An AI learns from photos tagged “cat” or “dog” by people. Wrong tags teach it wrong.
- KEY IDEA** Find the **true tags** and **whom to trust** together: each one helps find the other.
- USED FOR** Cleaning crowd-sourced labels before training.

Trouble 1: people disagree

Trouble: three people tag the same photos, and **disagree**.

	photo 1	photo 2
person 1	cat	dog
person 2	cat	dog
person 3	dog	dog
true label	?	?

1. Photo 1: two say **cat**, one says **dog**.

2. A vote says cat, but it trusts **everyone** **equally**.

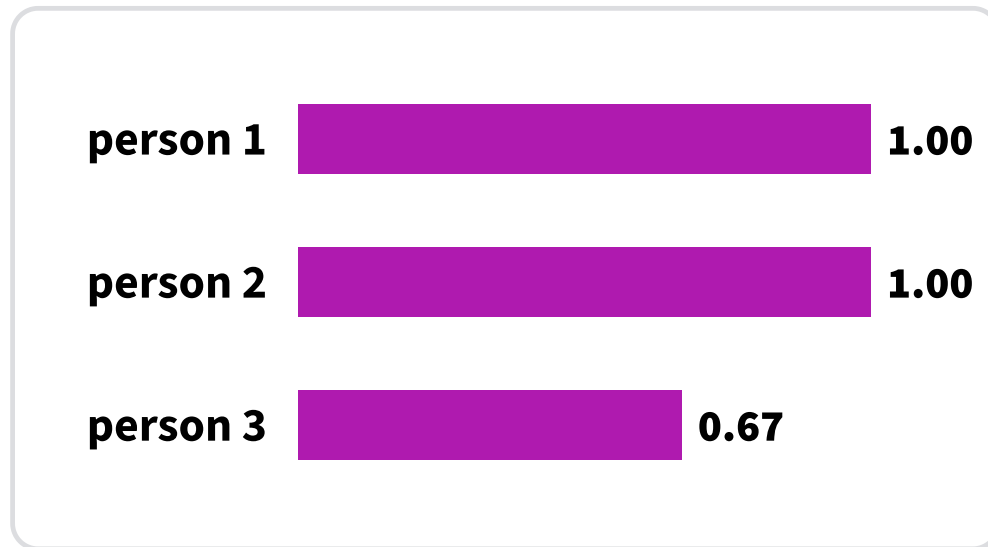
Q: What if person 3 is careless?

A: Then their vote should count less. But who is careless?

Trouble 1: people disagree

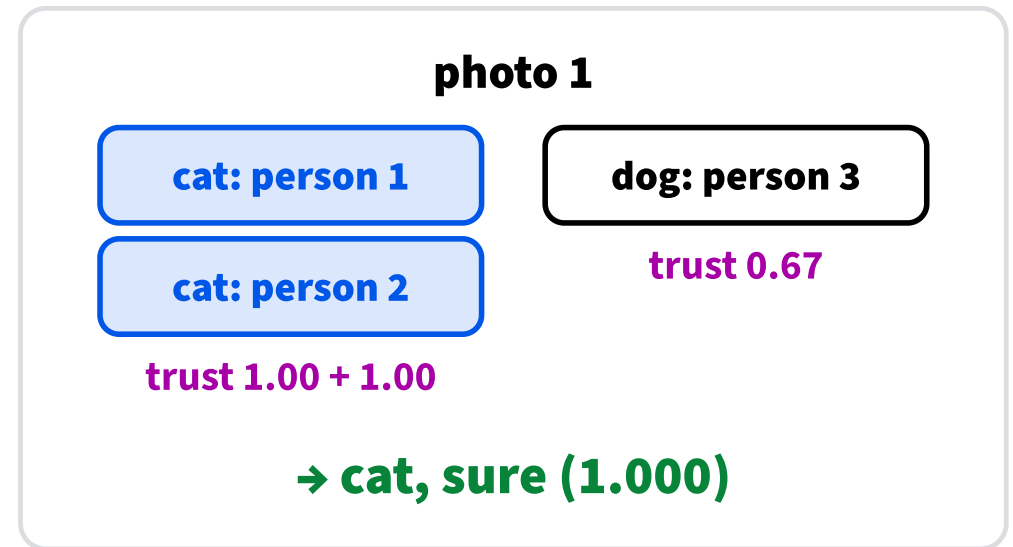
Main idea: learn whom to trust; then let trusted votes count more.

① Learn trust: EM



trust is learned from agreement alone

② Use trust: a weighted vote



a clean tag for the training set

Read: EM is phase ①: it learns trust and the tags together. Phase ② gives the clean tags.

Q: Did anyone tell EM the true tags? A: No. Trust comes only from who agrees with whom.

Trouble 1: people disagree

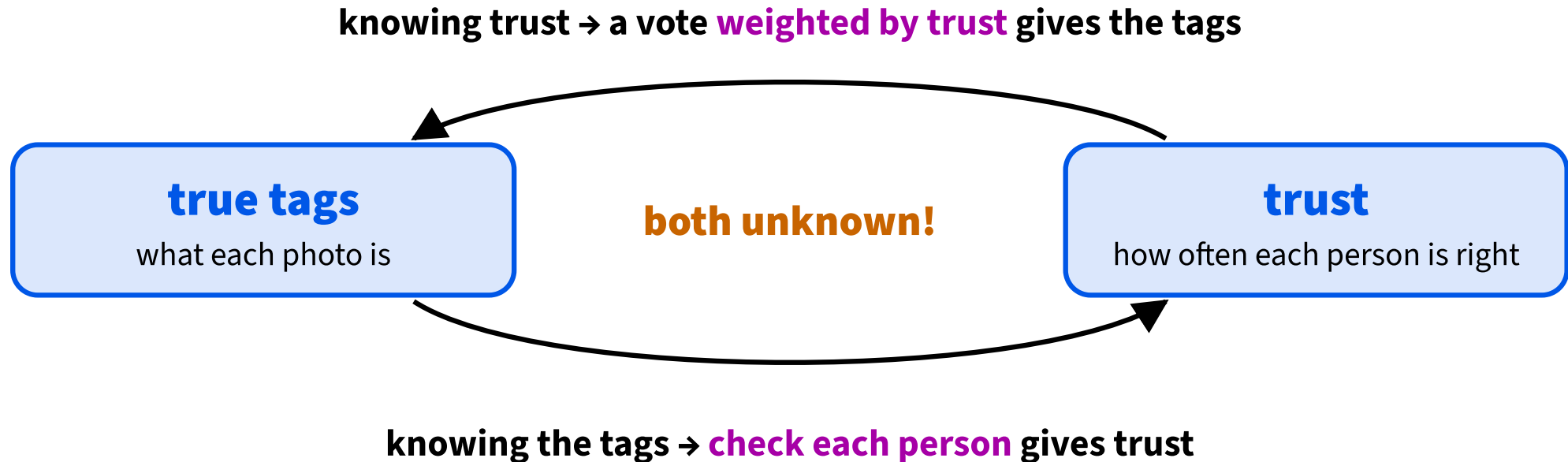
Scenario: the same EM as Lecture 5, in new clothes.

	Lecture 5: two coins (p. 41)	Crowd labels
record	two flips, e.g. HH	three people's tags on one photo
hidden coin	which coin: A or B	the true tag: cat or dog
θ	heads chances θ_A, θ_B	each person's trust
E-step	soft label for coin A (Bayes' rule)	chance the photo is a cat
M-step	weighted heads $\div$ weighted flips	weighted right tags $\div$ photos

Read: the same Bayes' rule and the same counting; “heads” just becomes “agrees with the true tag”.

Trouble 1: people disagree

Trouble: a chicken-and-egg problem.



Read: each unknown would give the other, but we have **neither**. Same as Lecture 5: the true tag is the hidden coin; trust is θ .

Q: Where do we start? A: Guess trust, then go round the loop: that is EM.

Trouble 1: people disagree

Fix: go round the loop. One round by hand:

START Trust everyone the same: **0.8**.

① **TAGS** Photo 1: cat, cat, dog → **80% cat**. Two votes beat one (**Lab 5, p. 62**).

② **TRUST** Person 3 went against the others on 2 of 6 photos → trust **0.8** → **0.72**.

REPEAT Person 3's vote now counts less, so the tags get surer, and person 3's trust settles at **0.67**.

Read: nobody told EM who is careless; it found out from the **disagreements**.

Trouble 1: people disagree

Why it works: better than a vote, no truth needed, one price.

BETTER

Each tag counts $\ln \frac{a}{1-a}$ votes, a = trust. One person at 0.95 (2.9 votes) outweighs two at 0.6 (0.4 each).

NO TRUTH

Trust is learned from **agreement** alone, and each round the likelihood rises or stays (Lecture 5, Theorem 4, p. 49).

PRICE

If most people are careless, EM trusts the wrong ones. Remedy: start from the majority vote.

Read: the classic way to merge crowd labels [Dawid and Skene, 1979].

Trouble 1: people disagree

Result: after a few rounds, EM knows **whom to trust**.

round 8 • ② score each person

	person 1	person 2	person 3	what is the photo?
photo 1	cat	cat	dog	cat 1.000
photo 2	dog	dog	dog	dog 1.000
photo 3	cat	cat	cat	cat 1.000
photo 4	dog	dog	cat	dog 1.000
photo 5	cat	cat	cat	cat 1.000
photo 6	dog	dog	dog	dog 1.000

person 3 matches the guesses on photos 1 ... 6:

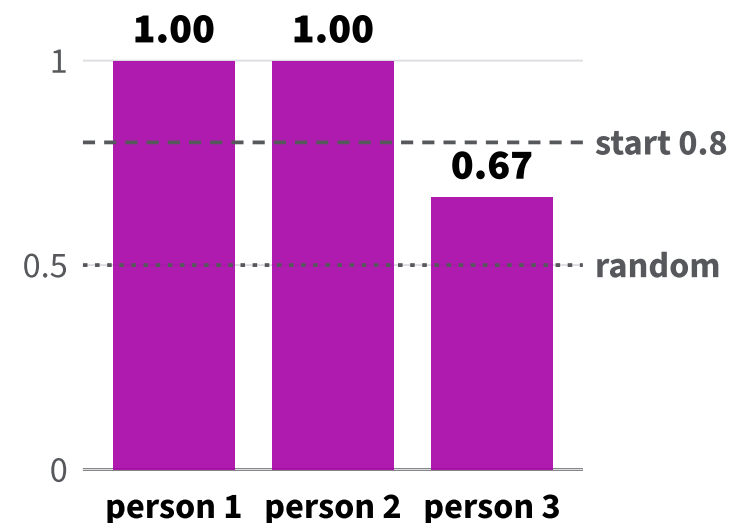
$$(0.000 + 1.000 + 1.000 + 0.000 + 1.000 + 1.000) \div 6 = 0.667$$

Watch: each round does ① then ②. The photos become sure (blue), person 3's trust falls to **0.67**, persons 1 and 2 rise to 1.

Q: Why does person 3 lose trust? A: They disagree with two people who agree with each other.

Try: person 3 guesses at random: trust 0.5, their tags stop counting. This is **Dawid-Skene EM** [Dawid and Skene, 1979].

trust: how often each person is right



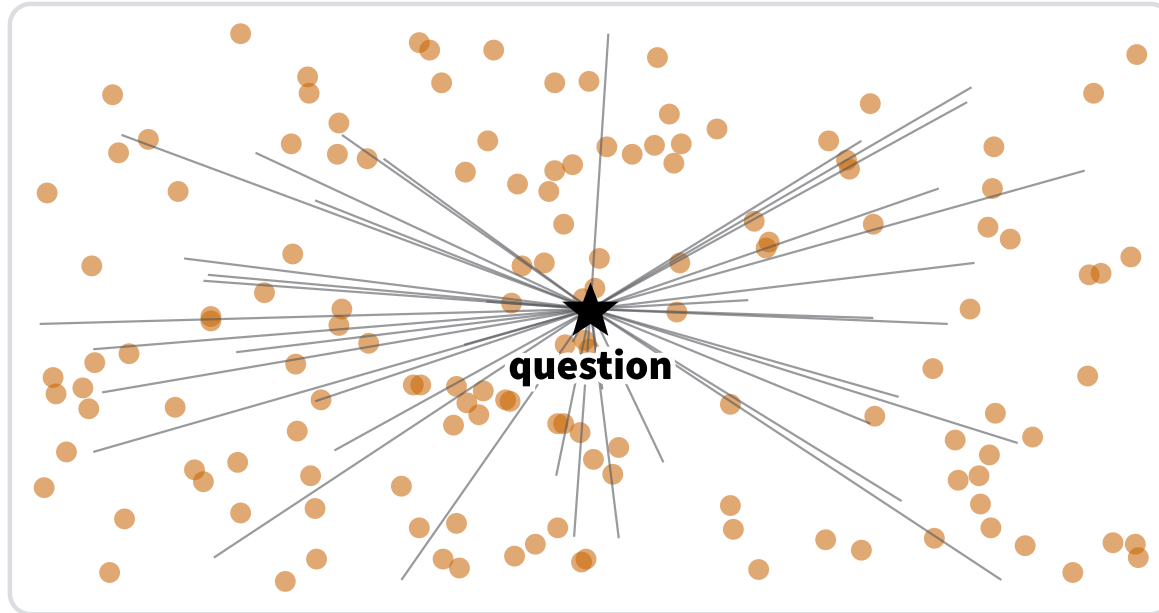
persons 1 and 2 always agree:
EM ends up trusting them fully

Trouble 2 in three sentences

- BACKGROUND** Before answering, a chatbot looks up texts in a library of millions. Each text is stored as a **point**: close points, similar meaning.
- KEY IDEA** Sort the texts onto **shelves** once, with EM's hard-label version (**k-means**); each question then searches **one shelf**.
- USED FOR** Vector search behind chatbots [Jégou, Douze and Schmid, 2011; Lewis et al., 2020].

Trouble 2: too much data

Trouble: each question must find its **nearest** text.



every dot is a text: compare with all of them?

1. A question becomes a point too.
2. Plain search compares it with **every** text.
3. A million texts, thousands of questions a second: **too slow**.

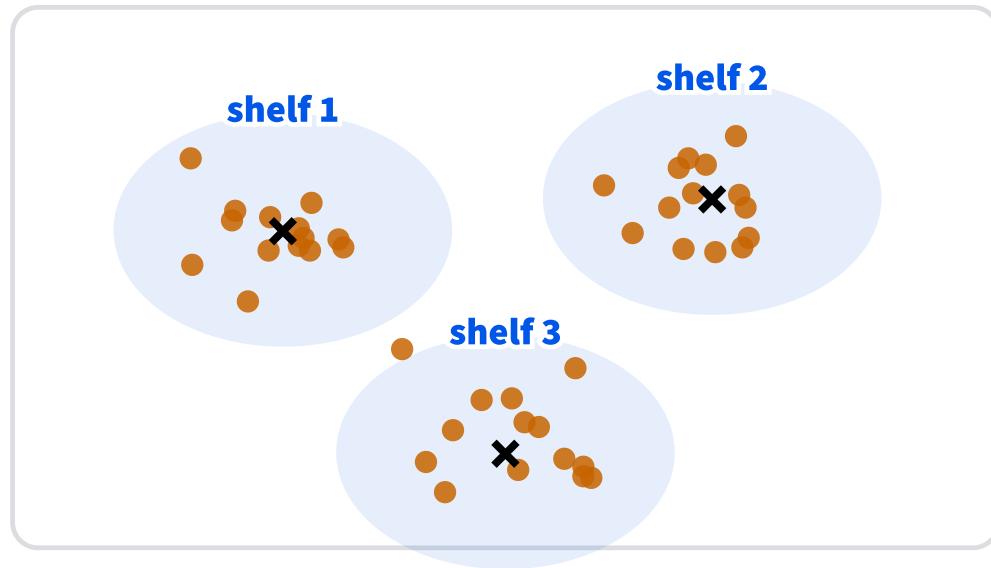
Q: What would a librarian do?

A: Shelf books by topic, then search one shelf.

Trouble 2: too much data

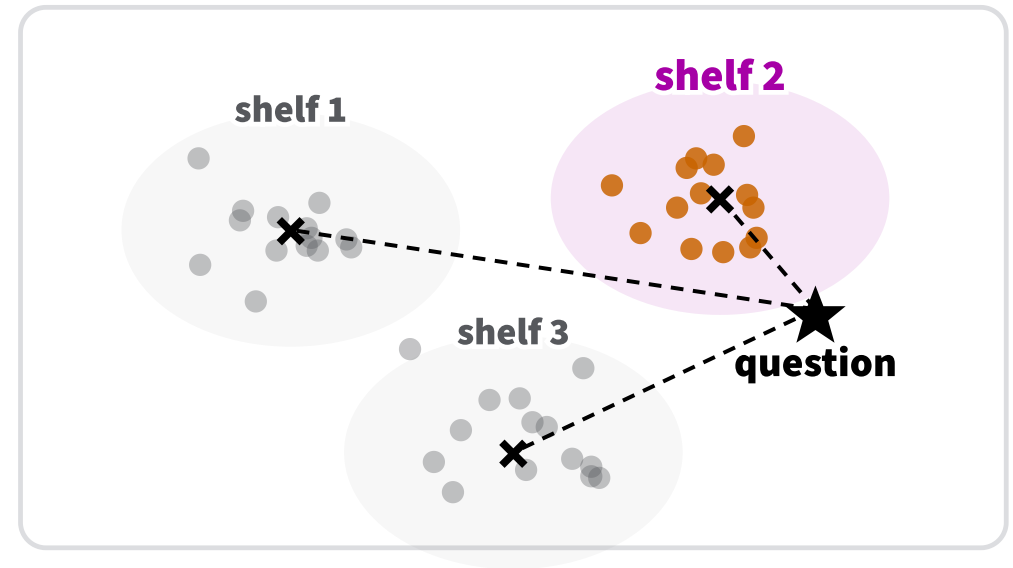
Main idea: sort the texts onto shelves once; then search **one shelf**.

① **Build, once: k-means**



every text goes on its nearest shelf

② **Search, every question**



3 centres, then **one shelf** only

Read: EM is phase ①: k-means is EM with hard labels. Phase ② is where the time is saved.

Q: Is the search itself EM? A: No. EM builds the shelves; every search reuses them.

Trouble 2: too much data

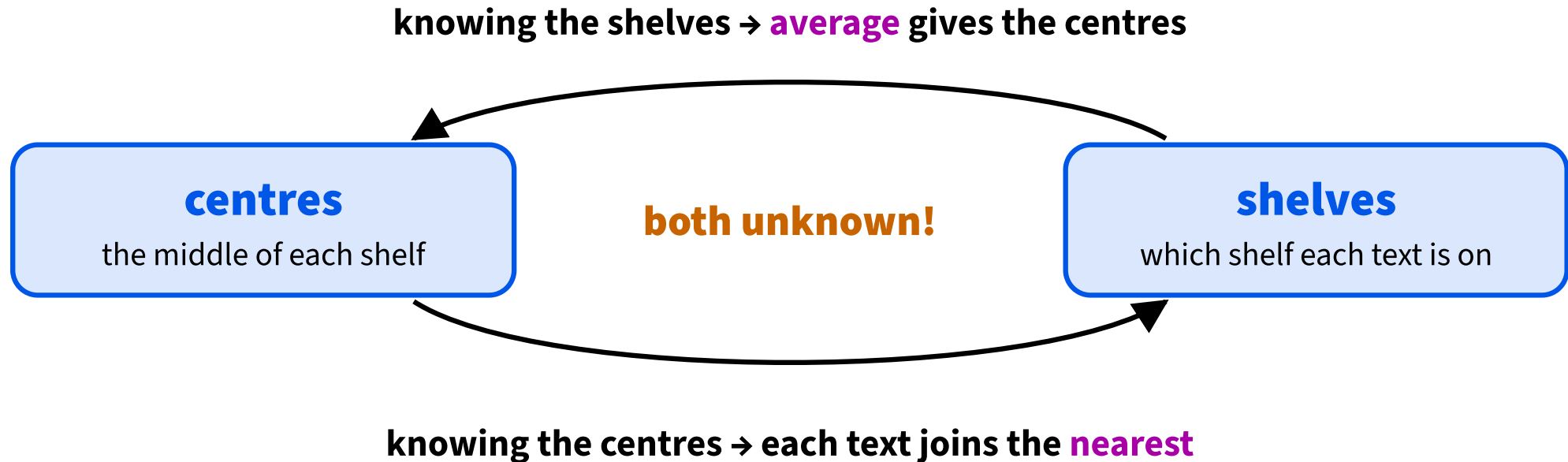
Scenario: phase ① is Lecture 5's EM, with hard labels.

	Lecture 5, Section 5 (p. 55)	Chatbot search
record	a number: 0, 1, 4, 5	a text, stored as a point
hidden coin	which bell made it	which shelf (topic) it is on
θ	the bell centres	the shelf centres
E-step	a share for each centre	nearest centre only (hard)
M-step	weighted average	plain average

Read: Lecture 5, Theorem 5 (p. 60): shrink every bell to width zero, and EM becomes the right column: **k-means**.

Trouble 2: too much data

Fix: shelf texts by topic. But that is **chicken and egg**.



Read: the same loop as EM. Start from guessed centres and go round: that is **k-means**.

Q: Why hard labels, not EM's soft shares? A: A text must sit on one shelf; “0.7 here, 0.3 there” cannot be filed.

Trouble 2: too much data

Fix: go round the loop. One round by hand:

START

Six texts at 1, 2, 3, 10, 11, 12. Guess centres **2** and **3** (a bad guess).

① E-STEP

Each text joins the **nearer** centre: {1, 2} and {3, 10, 11, 12}.

② M-STEP

Each centre moves to its texts' **average**: **1.5** and **9**.

REPEAT

Text 3 switches shelf; centres **2** and **11**; nothing moves: done.

SEARCH

A question at 10.4: 2 centres → shelf {10, 11, 12} → **5 distances, not 6**.

Trouble 2: too much data

Why it works: fast, usually right, and a known price.

FAST A question checks K centres + one shelf of about N/K texts. A million texts, 1,000 shelves: 2,000 distances, not 1,000,000: **500 times fewer**.

RIGHT k-means never increases the total squared distance from texts to their centres (Lecture 5, p. 59), so shelves are **tight**: the nearest text is usually on the question's shelf.

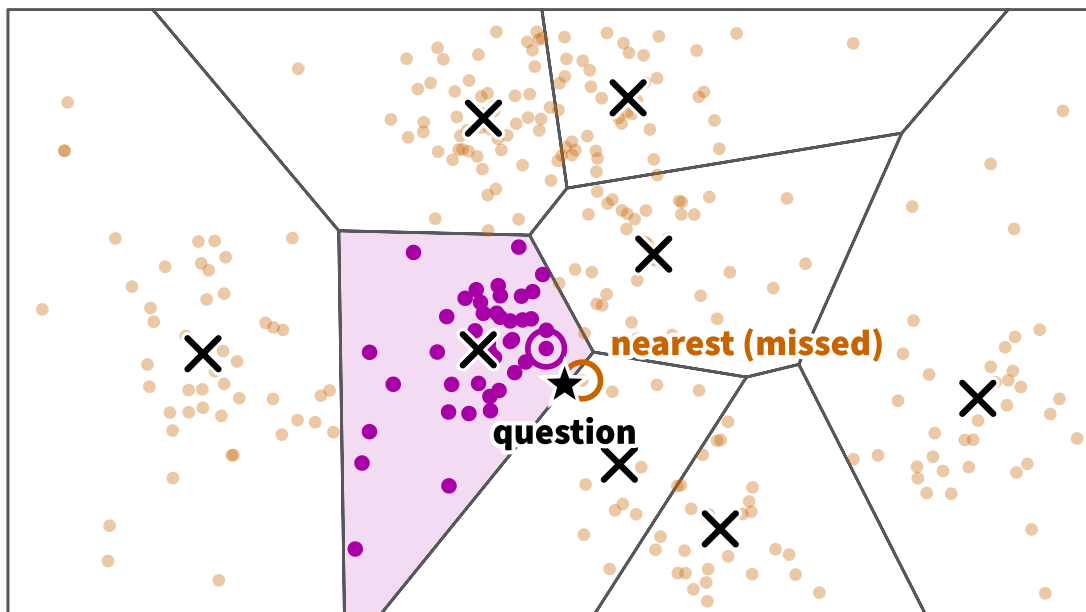
PRICE A neighbour just across a border is **missed**. Remedy: search the 2 or 3 nearest shelves, a step back toward EM's soft shares.

Read: a little accuracy for a lot of speed: this is how vector databases search [Jégou, Douze and Schmid, 2011].

Trouble 2: too much data

Result: search **one shelf (cell)**: 49 distances instead of 300.

texts as vectors (orange) · centres (black x)



search one cell (8 cells)

every text: 300

one cell: 8 + 41 = 49

nearest text: missed x

magenta ring: best in the cell, not the nearest

over 50 questions: 4% of nearest texts missed, about 50 distance computations each, not 300

Watch: the centres settle, then the question searches only its own cell. Sometimes the nearest text sits just across a border: **missed** (4%).

Q: Why are misses rare? A: Shelves are tight: the nearest text is usually near the same centre.

Try: more cells: faster, but more **misses**. This is **k-means search** [Lloyd, 1982].

Outline

Learning what we never see (Lecture 5)

Troubles 1–2: cleaning and searching data

► **Trouble 3: inside image models**

Conclusion and your turn

After class: Programs 4–5

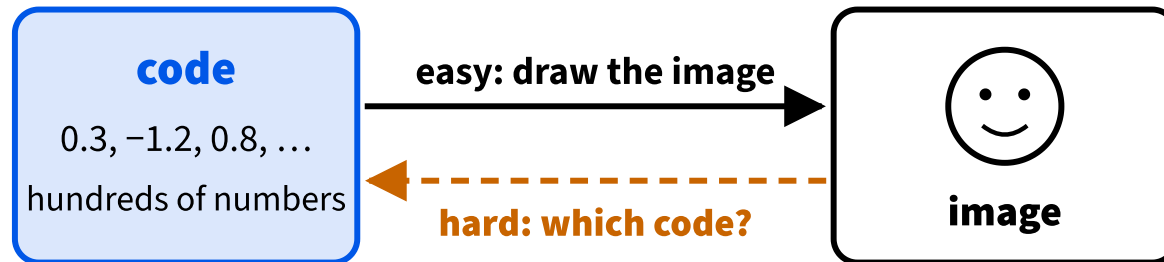
Next: inside an image model.

Trouble 3 in three sentences

- BACKGROUND** An image model draws each image from a hidden **code**: hundreds of numbers (pose, colour, ...).
- KEY IDEA** Train a second network, the **encoder**, to guess each image's code in one pass.
- USED FOR** Image models that compress, generate and edit.

Trouble 3: the code is many numbers

Trouble: to learn, the model must know each image's **code**.



endless codes could draw it

1. EM's guess step asks: for this image, how likely is each code?

2. Two coins: 2 options. Codes: **endless** options, no formula.

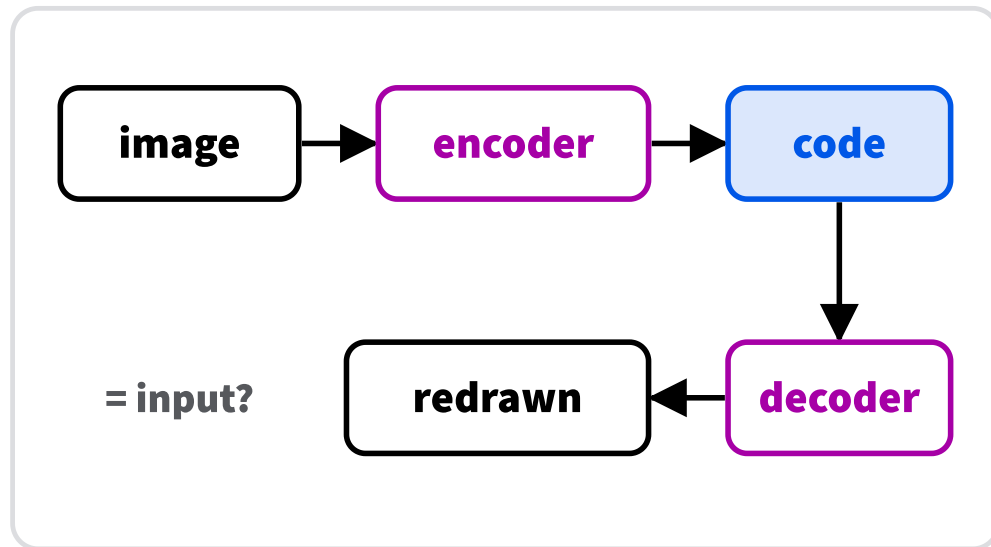
Q: Why not try many codes?

A: Hundreds of numbers: far too many combinations.

Trouble 3: the code is many numbers

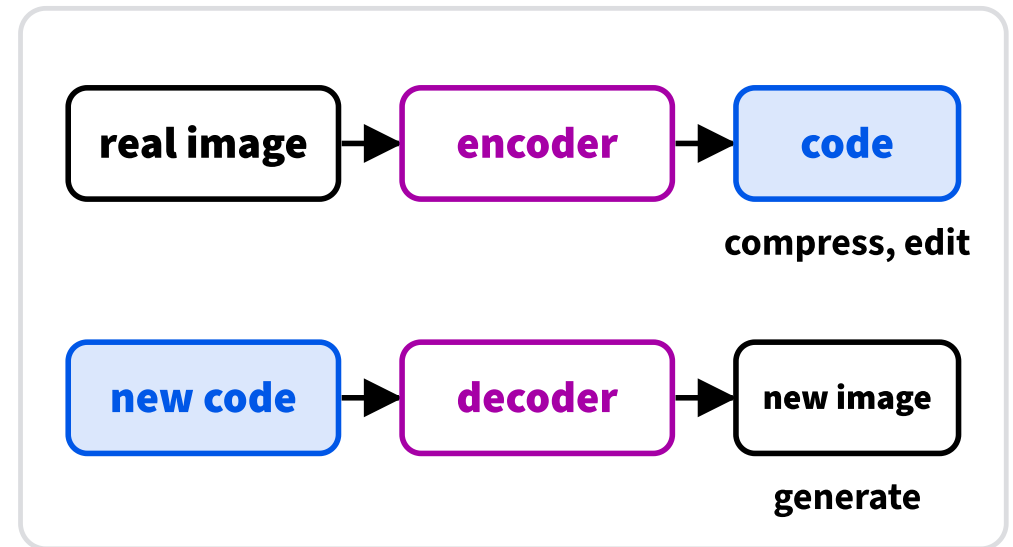
Main idea: train a fast guesser of the code, together with the model.

① Train both together: EM



raise the floor: guess the code, redraw the image

② Use them



compress and edit real images; make new ones

Read: EM is phase ①: the encoder is the E-step's guess; training the decoder is the M-step. Phase ② is what image models do.

Q: Why keep the encoder after training? A: To get a real image's code in one pass, to compress or edit it.

Trouble 3: the code is many numbers

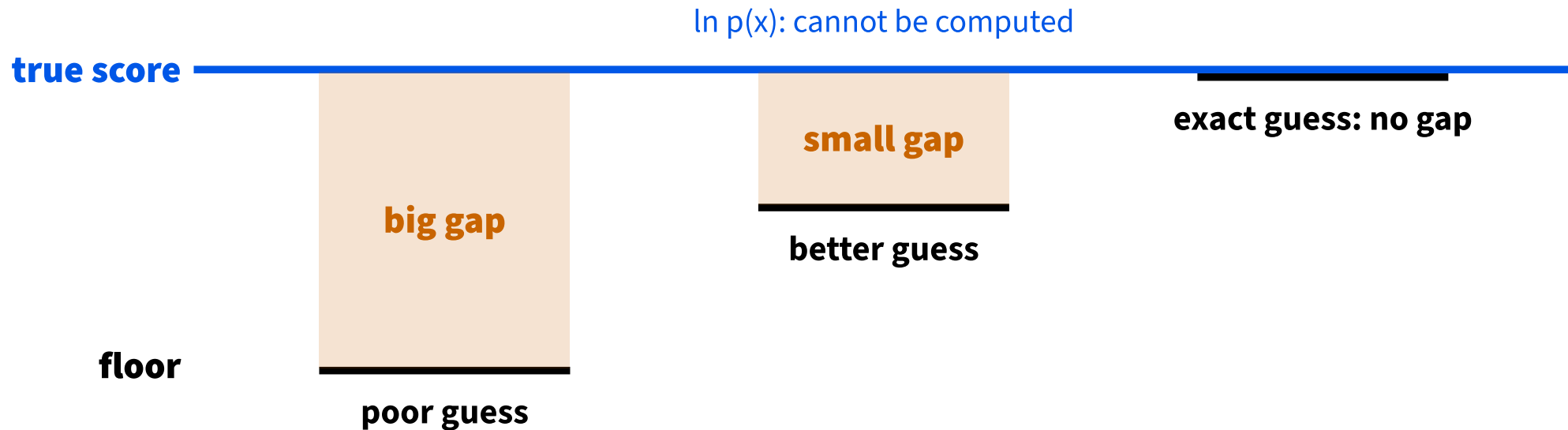
Scenario: the same EM as Lecture 5, in new clothes.

	Lecture 5: two coins (p. 41)	Image model
record	two flips, e.g. HH	an image
hidden coin	which coin: A or B	the code: hundreds of numbers
θ	heads chances	the decoder's weights
E-step	exact soft label (Bayes' rule)	the encoder's guess q
M-step	count and divide	raise the floor by gradient steps

Read: Lecture 5, Lemma 1 (p. 47): $\ln p(x) = \text{floor (bound)} + \text{gap}$. EM makes the gap zero; here no formula exists, so the encoder makes it small.

Trouble 3: the code is many numbers

Trouble: the exact answer is out of reach.



Read: we cannot compute the true score, but **every guess gives a floor** under it. The better the guess, the smaller the **gap**.

Q: So what do we train? A: Raise the floor: it trains the model and the guesser together.

Trouble 3: the code is many numbers

Fix: go round the loop. One round by hand:

- START** A coin stands in for an image. On HH the exact soft label is **0.9**; the guesser says 0.5: gap **0.51**.
- ① **TRAIN** The guess moves to 0.8: gap **0.04** (**Lab 5, p. 63**).
- ② **TRAIN** The guess reaches 0.9: gap **0**; the floor touches the true score.
- IMAGES** Nobody knows the exact answer; training **finds** it by shrinking the gap, and then one pass gives the guess.

Read: the encoder replaces an impossible search by a fast guess.

Trouble 3: the code is many numbers

Why it works: fast, always safe, one price.

FAST One pass of the encoder per image, instead of a search over endless codes.

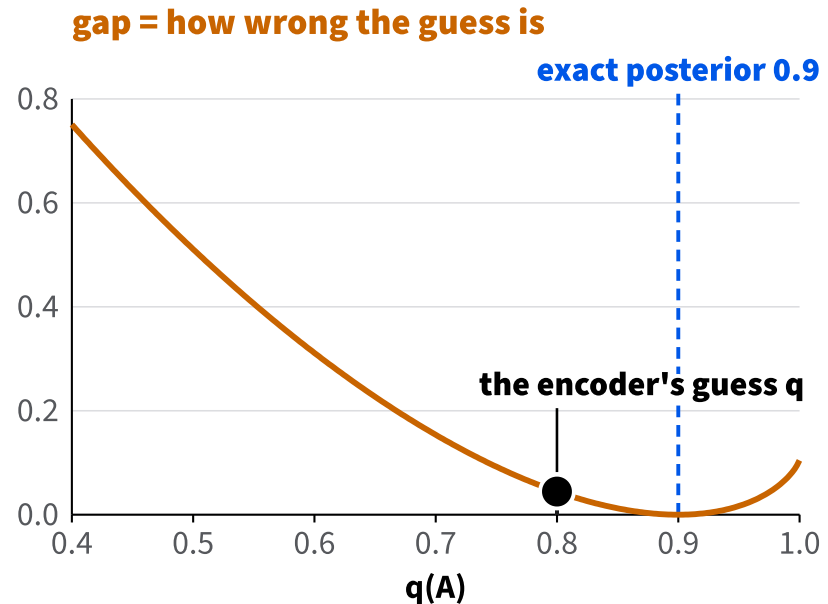
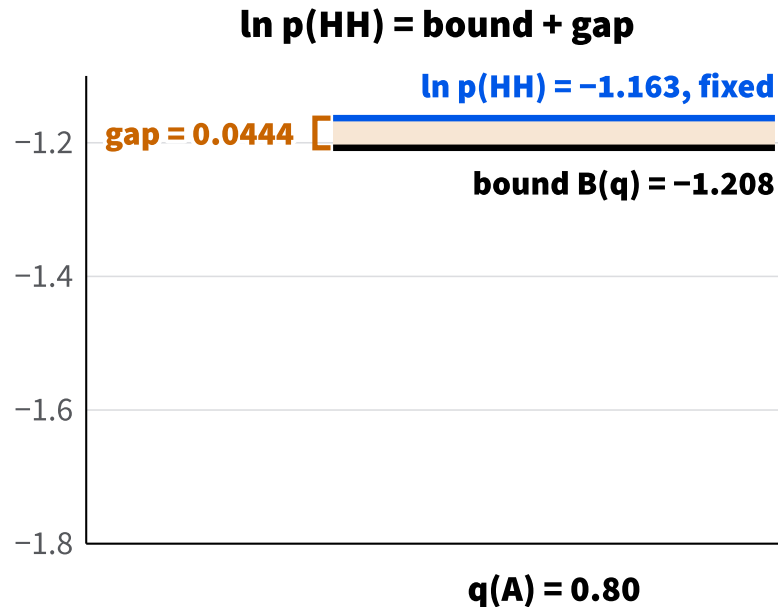
SAFE Every guess gives a floor under the true score (Lecture 5, Lemma 1, p. 47); raising it improves the model or the guess, never neither.

PRICE The gap is rarely zero: one encoder for all images guesses less well than a separate guess per image [Cremer et al., 2018].

Read: this floor-raising trick trains the hidden codes of modern image models.

Trouble 3: the code is many numbers

Result: training slides the guess down the **gap** curve.



Watch: on a coin we can compute the exact 0.9, since there are only two options. For an image nobody can: the encoder must **find** it, shrinking the orange gap step by step.

Q: Then why show 0.9? A: To see the target: the gap is zero only there.

Try: set the guess to 0.99, surer than the truth (0.9): the gap grows again, to 0.07. **Too sure is wrong too.** This is the **variational autoencoder** [Kingma and Welling, 2014].

Outline

Learning what we never see (Lecture 5)

Troubles 1–2: cleaning and searching data

Trouble 3: inside image models

► **Conclusion and your turn**

After class: Programs 4–5

Finally: all five fixes, and your turn.

Conclusion

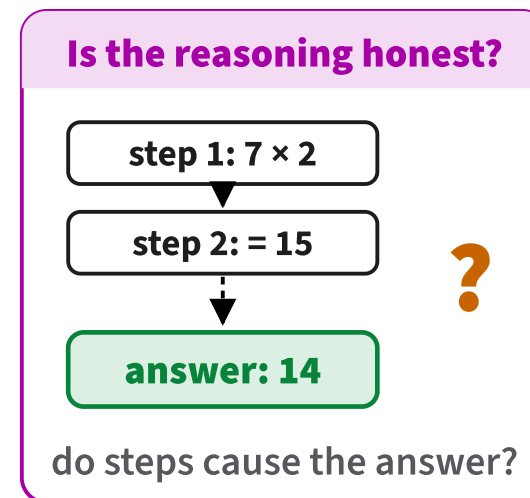
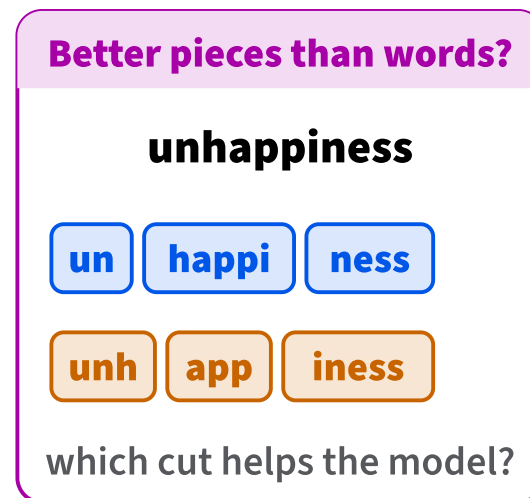
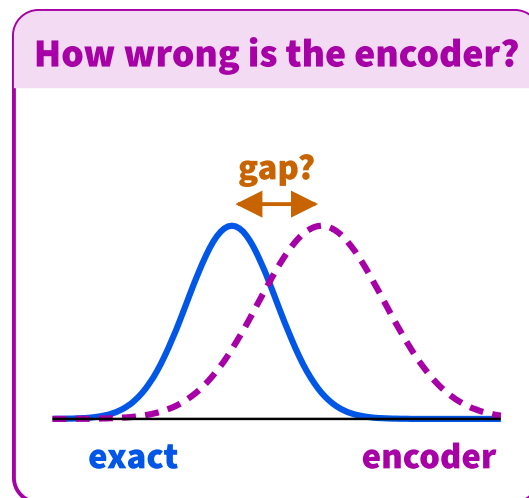
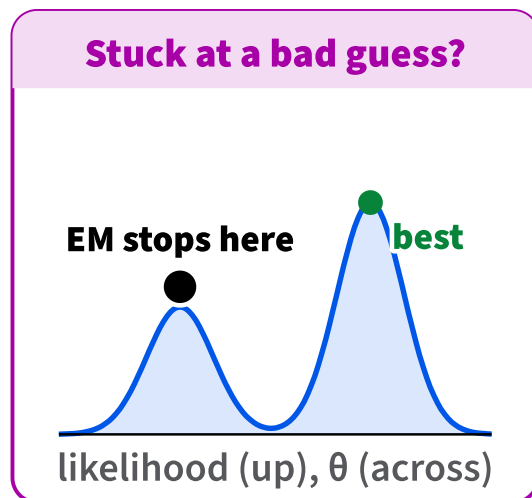
One loop, five small changes.

	Trouble	Fix	Used for
1	people disagree on labels	trust who agrees	clean training labels
2	too much data	search one cell	search for chatbots
3	the hidden part is many numbers	a fast guesser	image models
4	nobody marks where to cut words	weigh every cut	tokenizers · program
5	the reasoning is never written	keep what works	reasoning models · program

Read: every fix keeps **guess**, then **count**.

Conclusion

- ▶ We can **learn what we never see**: guess, then count (expectation-maximization).
- ▶ Four **open problems**:



Project seeds: Jin et al. (2016), Cremer et al. (2018), Pagnoni et al. (2025), Turpin et al. (2023).

Your turn

Predict, then try.

In these slides

- ▶ another start (**Lab 5, p. 5**): still 0.789?
- ▶ a careless labeller (**Lab 5, p. 17**)
- ▶ more cells (**Lab 5, p. 25**)
- ▶ a guess surer than 0.9 (**Lab 5, p. 34**)

After class: `lab5_explore.py`

1. Two coins from $\theta = (0.6, 0.6)$: where to?
2. Eight photos: how much is person 3 trusted?
3. k-means: how fast, how many misses?
4. **Program 4**: a word cutter (**Lab 5, p. 50**)
5. **Program 5**: a tiny reasoner (**Lab 5, p. 58**)

Course site → Week 5 → Lab files. One or two lines each.

Tip: write your guess first; surprises teach.

Selected References I

Start here: Dempster, Laird and Rubin (1977), where EM got its name.



Cremer, C., Li, X., and Duvenaud, D. (2018).

Inference suboptimality in variational autoencoders.

In Proceedings of the 35th International Conference on Machine Learning, PMLR 80:1078–1086.



Dawid, A. P. and Skene, A. M. (1979).

Maximum likelihood estimation of observer error-rates using the EM algorithm.

Journal of the Royal Statistical Society, Series C (Applied Statistics), 28(1):20–28.



Dempster, A. P., Laird, N. M., and Rubin, D. B. (1977).

Maximum likelihood from incomplete data via the EM algorithm.

Journal of the Royal Statistical Society, Series B (Methodological), 39(1):1–38.



Jégou, H., Douze, M., and Schmid, C. (2011).

Product quantization for nearest neighbor search.

IEEE Transactions on Pattern Analysis and Machine Intelligence, 33(1):117–128.



Jin, C., Zhang, Y., Balakrishnan, S., Wainwright, M. J., and Jordan, M. I. (2016).

Local maxima in the likelihood of Gaussian mixture models: structural results and algorithmic consequences.

In Advances in Neural Information Processing Systems 29, pages 4116–4124.

Selected References II

Entry points: Kingma and Welling (2014); Kudo (2018).



Kingma, D. P. and Welling, M. (2014).

Auto-encoding variational Bayes.

In 2nd International Conference on Learning Representations; arXiv:1312.6114.



Kudo, T. (2018).

Subword regularization: improving neural network translation models with multiple subword candidates.

In Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics, pages 66–75.



Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W., Rocktäschel, T., Riedel, S., and Kiela, D. (2020).

Retrieval-augmented generation for knowledge-intensive natural language processing tasks.

In Advances in Neural Information Processing Systems 33, pages 9459–9474.



Lloyd, S. P. (1982).

Least squares quantization in pulse-code modulation.

IEEE Transactions on Information Theory, 28(2):129–137.

Selected References III

A short read: Zelikman et al. (2022), a model teaching itself to reason.



Pagnoni, A., Pasunuru, R., Rodriguez, P., Nguyen, J., Muller, B., Li, M., Zhou, C., Yu, L., Weston, J., Zettlemoyer, L., Ghosh, G., Lewis, M., Holtzman, A., and Iyer, S. (2025).

Byte latent transformer: patches scale better than tokens.

In Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), pages 9238–9258.



Phan, D., Hoffman, M. D., Dohan, D., Douglas, S., Le, T. A., Parisi, A., Sountsov, P., Sutton, C., Vikram, S., and Sauros, R. A. (2023).

Training chain-of-thought via latent-variable inference.

In Advances in Neural Information Processing Systems 36.



Turpin, M., Michael, J., Perez, E., and Bowman, S. R. (2023).

Language models don't always say what they think: unfaithful explanations in chain-of-thought prompting.

In Advances in Neural Information Processing Systems 36.



Zelikman, E., Wu, Y., Mu, J., and Goodman, N. D. (2022).

STaR: bootstrapping reasoning with reasoning.

In Advances in Neural Information Processing Systems 35, pages 15476–15488.

Thank you!

Next: Lecture 6, when even the guess must be approximated

After class: Programs 4 and 5 follow, then the backup slides.

Outline

Learning what we never see (Lecture 5)

Troubles 1–2: cleaning and searching data

Trouble 3: inside image models

Conclusion and your turn

► **After class: Programs 4–5**

Two more troubles, same structure, for you to read and program.

Program 4 in three sentences

- BACKGROUND** A language model reads text in pieces (**tokens**): un|happi|ness. Good pieces make it faster.
- KEY IDEA** Learn the pieces from raw text: EM weighs **every cut** of every word.
- USED FOR** Tokenizers of T5 and ALBERT [Kudo, 2018].

Program 4: where to cut words

Scenario: the same EM as Lecture 5, in new clothes.

	Lecture 5: two coins (p. 41)	Tokenizer
record	two flips, e.g. HH	a word, e.g. “noon”
hidden coin	which coin: A or B	how the word is cut
θ	heads chances	each piece's chance
E-step	soft label of each coin	share of each cut
M-step	weighted heads $\div$ weighted flips	weighted piece counts $\div$ all pieces

Read: many cuts per word, but a running total adds them letter by letter (forward–backward, Lecture 5, p. 71).

Program 4: where to cut words

Trouble: nobody marks where to **cut**.

n o o n

n o o n	?
n o on	?
n oo n	?
no o n	?
no on	?

1. With the pieces n, o, no, on, oo, “noon” has **five** cuts.

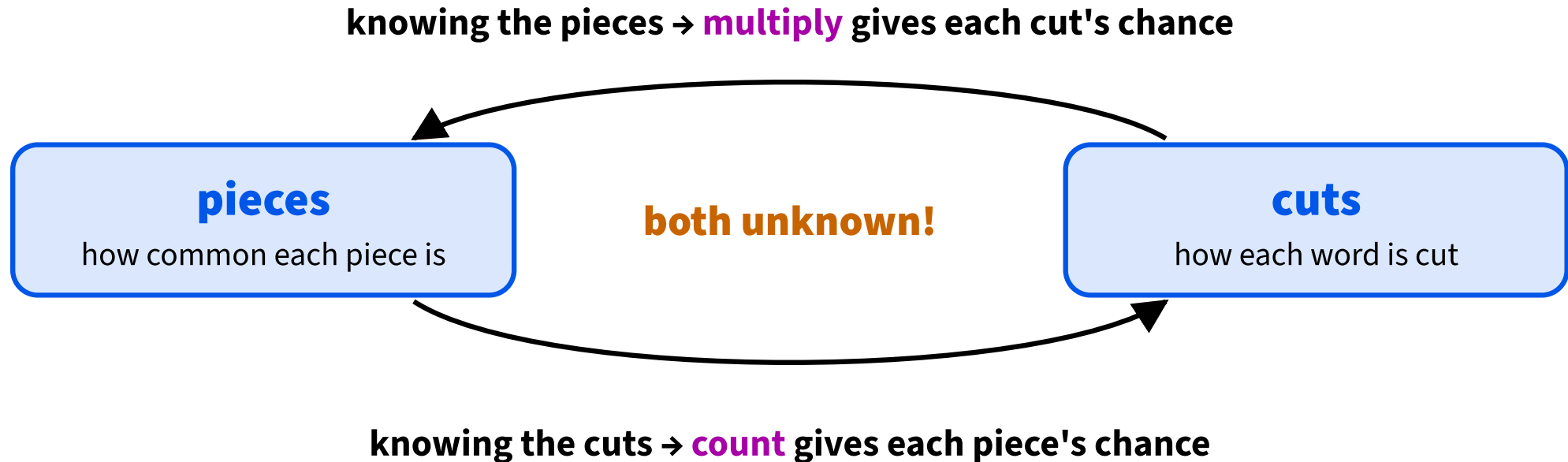
2. Which cut is right? Nobody tells us.

Q: Why not cut at every letter?

A: Four pieces where two would do: slower, and letters carry no meaning.

Program 4: where to cut words

Trouble: a **chicken-and-egg** problem again.



Read: the cut is the hidden coin; a piece's chance is θ .

Q: Where do we start? A: Make every piece equally likely, then go round the loop.

Program 4: where to cut words

Fix: go round the loop. One round by hand:

START Every piece equally likely: **1/5** each.

① CUTS A cut's chance = its pieces multiplied: no|on = $1/25$, three pieces = $1/125$. So no|on gets **0.61**.

② PIECES Count pieces, weighted by the cuts: **no** and **on** gain.

REPEAT no|on grows surer each round: **0.999** by round 4.

Read: fewer pieces means fewer small numbers multiplied, so the natural cut wins.

Program 4: where to cut words

Why it works: it improves, it is fast, one price.

IMPROVES

The chance of the text never drops (Lecture 5, Theorem 4, p. 49): common pieces such as **no** and **on** win on their own.

FAST

The running total grows with the word's length, not with its up to 524,288 cuts.

PRICE

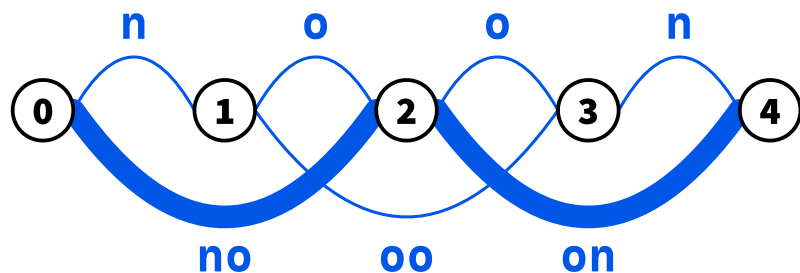
A poor start can stick (favour “oo”: stuck at $n|oo|n$), and the list of candidate pieces must be chosen first.

Read: the Unigram tokenizer [Kudo, 2018], in the SentencePiece software.

Program 4: where to cut words

Result: EM finds the natural cut **no|on**.

every path 0 → 4 is a cut of “noon”



thicker arc = likelier piece

**the cut no|on becomes
almost certain (0.999)**

E-step: share of each cut

= its chance ÷ chance of “noon”

n|o|o|n | 0.000

n|o|on | 0.001

n|oo|n | 0.000

no|o|n | 0.001

no|on | 0.999

M-step: piece probability

= expected count ÷ 2.00 pieces

n | 0.001

o | 0.001

no | 0.499

on | 0.499

oo | 0.000

Watch: round by round **no** and **on** gain, and no|on takes all the weight.

Q: Why not list every cut? A: A 20-letter word has up to 524,288; a running total adds them letter by letter.

Try: favour “oo”: EM gets stuck. This is the **Unigram tokenizer** [Kudo, 2018].

Program 4: a word cutter

Program: build it yourself: `python lab5_explore.py` 4

1. List every cut of “noon” from the pieces n, o, no, on, oo.
2. E-step: share of each cut = product of its pieces' chances $\div$ total.
3. M-step: piece chance = weighted count $\div$ all pieces. Repeat.

Check: all pieces equal at the start: no|on takes 0.61, then 0.999 by round 4.

Go further: replace the list of cuts by a running total, letter by letter (forward–backward), and get the same shares.

Program 5 in three sentences

- BACKGROUND** To teach reasoning we have questions and right answers, but **no worked steps**.
- KEY IDEA** Let the model write its own steps, **keep what reaches the right answer**, learn from it.
- USED FOR** Teaching language models to reason [Zelikman et al., 2022].

Program 5: the hidden reasoning

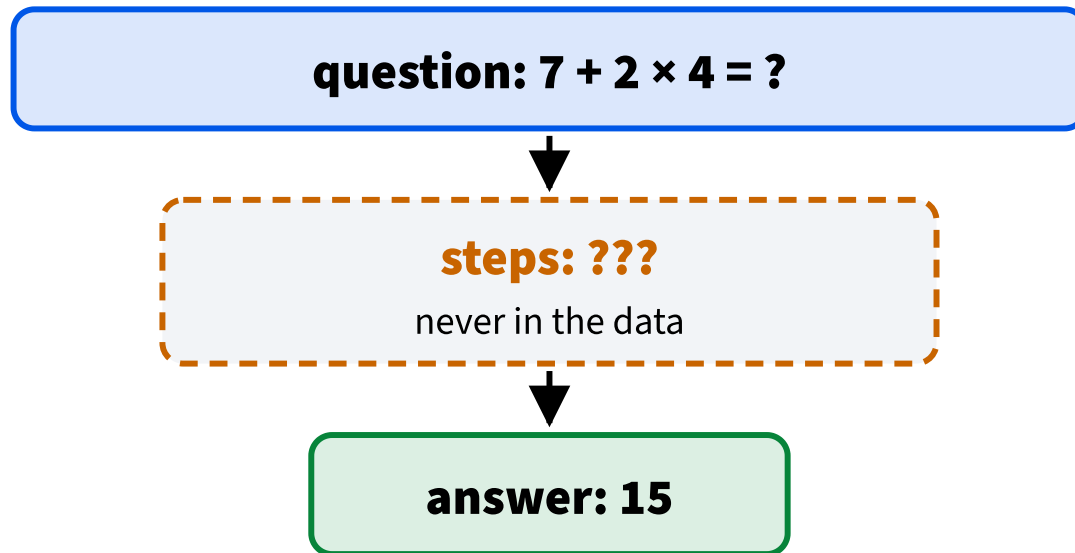
Scenario: the same EM as Lecture 5, in new clothes.

	Lecture 5: two coins (p. 41)	A model that reasons
record	two flips, e.g. HH	a question with its right answer
hidden coin	which coin: A or B	the worked steps
θ	heads chances	how often it reasons carefully
E-step	exact soft label	sample steps; keep those that reach the answer
M-step	count and divide	retrain on the kept steps

Read: too many possible steps to weigh them all, so we sample them: Monte Carlo EM (Lecture 5, Section 7, p. 88).

Program 5: the hidden reasoning

Trouble: the worked **steps** are missing.



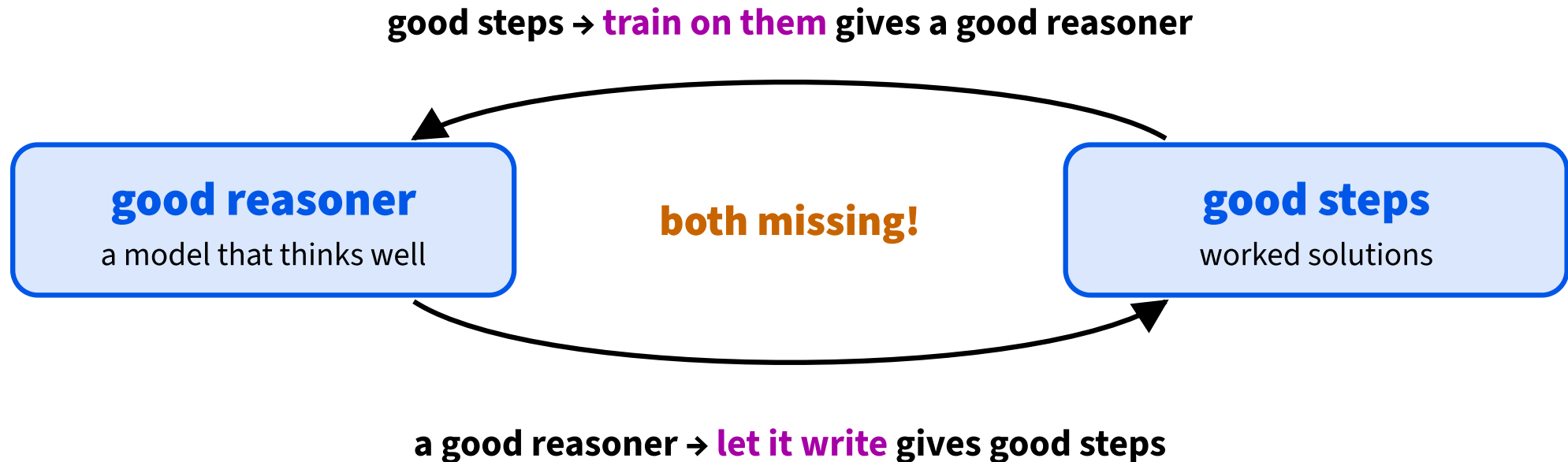
1. The data give the question and the answer.
2. The steps between are hidden, like the coin.

Q: Why not train on the answers alone?

A: The model learns to guess answers, not to reason.

Program 5: the hidden reasoning

Trouble: a chicken-and-egg problem again.



Read: the steps are the hidden coin; how often the model reasons carefully is θ .

Q: Where do we start? A: With the model we have, then go round the loop.

Program 5: the hidden reasoning

Fix: go round the loop. One round by hand:

- START** A tiny model reasons **carefully** half the time (right 3 in 4) or **hastily** (right 1 in 4).
- ① **WRITE** 8 tries: 4 careful give 3 right; 4 hasty give 1 right.
- ② **KEEP** Keep the 4 right ones: 3 careful, 1 hasty. Retrain: careful **3/4** (**Lab 5, p. 64**).
- REPEAT** Careful grows: $3/4 \rightarrow 0.9 \rightarrow$ nearly always.

Read: the right answer acts as a filter: it lets mostly careful steps through.

Program 5: the hidden reasoning

Why it works: it is the E-step, it is cheap, one price.

- E-STEP** Keeping only right answers samples steps **given the answer** (Lecture 5, p. 91): exactly the soft label (**Lab 5, p. 64**).
- CHEAP** Nobody writes the steps; questions with answers are enough.
- PRICE** A wrong chain can reach the right answer by luck and be kept; the steps may not be honest [Turpin et al., 2023].

Read: STaR [Zelikman et al., 2022] and its Monte Carlo EM reading [Phan et al., 2023].

Program 5: the hidden reasoning

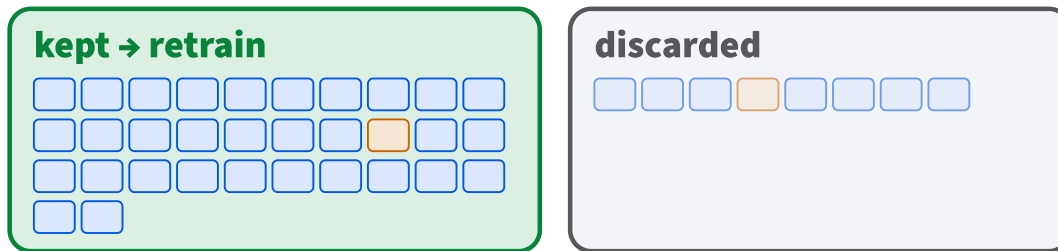
Result: careful reasoning takes over.

the model answers 40 questions

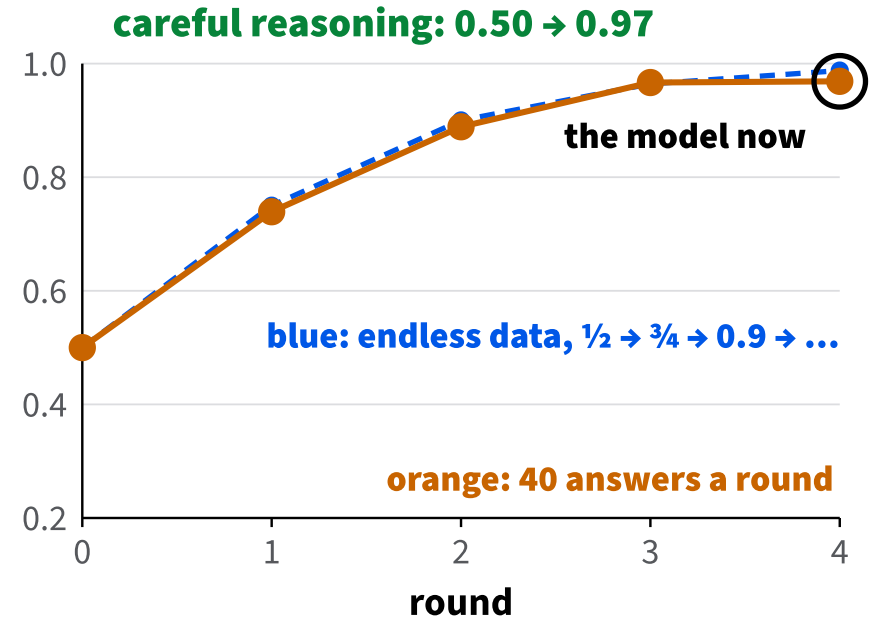
☐ careful: right $\frac{3}{4}$ ☐ hasty: right $\frac{1}{4}$

kept 32: 31 careful, 1 hasty

retrain: new $s = 31/32 = 0.969$



s = share of careful reasoning



Watch: each round the model writes 40 solutions, keeps the right ones and retrains: careful rises from 0.50 to 0.97.

Q: Why not keep every solution? A: Then nothing changes: careful stays at 1/2.

Try: keep all. This is STaR, a **Monte Carlo EM** [Zelikman et al., 2022; Phan et al., 2023].

Program 5: a tiny reasoner

Program: build it yourself: `python lab5_explore.py 5`

1. Each round, sample 1,000 answers: careful (right 3 in 4) or hasty (right 1 in 4).
2. E-step: keep only the right ones.
3. M-step: careful share = careful $\div$ kept. Repeat.

Check: careful rises 0.5 $\rightarrow$ about 0.73 $\rightarrow$ about 0.91; keep everything and it stays near 0.5.

Go further: let a hasty chain be right 1 in 2. Does keeping right answers still help?

Backup: one round of Lecture 5

One round from $\theta = (\frac{3}{4}, \frac{1}{4})$, coins picked $\frac{1}{2}$ each.

$$p(A \mid HH) = \frac{9/32}{9/32 + 1/32} = \frac{9}{9 + 1} = 0.9$$

$$p(A \mid HT) = \frac{3}{3 + 3} = 0.5, \quad p(A \mid TT) = \frac{1}{1 + 9} = 0.1$$

$$\text{heads for A} = 0.9 \cdot 2 + 0.5 \cdot 1 + 0.1 \cdot 0 = 2.3$$

$$\text{flips for A} = (0.9 + 0.5 + 0.1) \cdot 2 = 3$$

$$\theta_A = \frac{2.3}{3} = 0.767, \quad \theta_B = \frac{3 - 2.3}{6 - 3} = 0.233$$

Notice: heads $\div$ flips, each record weighted by its soft label.

E-step: $\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{3}{4} = 9/32$ against $\frac{1}{2} \cdot \frac{1}{4} \cdot \frac{1}{4}$

the same rule

soft label $\times$ heads

soft label $\times$ flips

M-step: count $\div$ total

Backup: one round of Lecture 5

EM in 11 lines of Python.

```
records = ["HH", "HT", "TT"]
a, b = 0.75, 0.25 # starting guess: heads chances of coins A and B
for r in range(1, 31):
    heads = flips = 0.0
    for x in records:
        h, t = x.count("H"), x.count("T")
        pa, pb = a**h * (1-a)**t, b**h * (1-b)**t # chance of the record from each coin
        s = pa / (pa + pb) # E-step: soft label for coin A
        heads += s * h; flips += s * (h + t) # count with the soft label
    a, b = heads / flips, (3 - heads) / (6 - flips) # M-step: expected heads / expected flips
    if r in (1, 2, 30): print(r, round(a, 3), round(b, 3))
1 0.767 0.233
2 0.777 0.223
30 0.789 0.211
```

Try: start from $a = b = 0.5$. What happens?

Backup: Trouble 1

One round on the six photos: every accuracy starts at 0.8, cat and dog $\frac{1}{2}$ each.

$$p(\text{cat} \mid \text{cat}, \text{cat}, \text{dog}) = \frac{0.064}{0.064 + 0.016} = 0.8$$

$$p(\text{dog} \mid \text{dog}, \text{dog}, \text{dog}) = \frac{0.256}{0.256 + 0.004} = 0.9846$$

$$a_1 = a_2 = \frac{0.8 + 0.8 + 4 \cdot 0.9846}{6} = 0.923$$

$$a_3 = \frac{0.2 + 0.2 + 4 \cdot 0.9846}{6} = 0.723$$

Notice: accuracy is counted like heads $\div$ flips, with “right” as heads.

E-step, photos 1 and 4

E-step, photos 2, 3, 5, 6: all agree

expected right $\div$ photos

person 3 is outvoted on photos 1 and 4

Backup: Trouble 3

The gap for the guess 0.8 on HH (exact: 0.9).

Lecture 5, Lemma 1 (p. 47): $\ln p(x) = \text{bound} + \text{gap}$; the gap is the Kullback–Leibler divergence.

$$\begin{aligned}\text{gap} &= 0.8 \cdot \ln \frac{0.8}{0.9} + 0.2 \cdot \ln \frac{0.2}{0.1} \\ &= 0.8 \cdot (-0.1178) + 0.2 \cdot 0.6931 \\ &= -0.0942 + 0.1386 = 0.0444\end{aligned}$$

Notice: at the exact guess 0.9 both logs are 0: no gap.

| guess against exact, for A and for B

| the two logarithms

| the bound is 0.0444 too low

Backup: Program 5

The tiny reasoner: careful = coin A, hasty = coin B, “right” = heads.

Careful: chance $\pi = \frac{1}{2}$, right $\frac{3}{4}$. **Hasty:** right $\frac{1}{4}$.

$$p(\text{careful and right}) = \frac{1}{2} \cdot \frac{3}{4} = \frac{3}{8}, \quad p(\text{hasty and right}) = \frac{1}{2} \cdot \frac{1}{4} = \frac{1}{8}$$

$$\text{new } \pi = \frac{3/8}{3/8 + 1/8} = 0.75$$

$$\text{keep all: } \pi = \frac{1/2}{1/2 + 1/2} = 0.5$$

sample, then check

share of careful among kept

nothing is learned

Notice: keeping = E-step (careful given right); retraining = M-step.